

MCU – SEED STUDIO XIAO ESP32–C6

U1
XIAO-ESP32-C6-SMD

GPIO0_A0_D0
GPIO1_A1_D1
GPIO2_A2_D2
GPIO21_D3
GPIO22_D4_SDA
GPIO23_D5_SCL
GPIO16_D6_TX
GPIO17_D7_RX
GPIO19_D8_SCK
GPIO20_D9_MISO
GPIO18_D10_MOSI

ESP_TX
ESP_RX

VBUS
3V3
GND
PWR_FLAG

1/2
1/3
8
9
10
11

Powered from the XIAO's own USB-C port (no board power input).
 ESP_TX = D3 / GPIO21 (uart_tx_pin)
 ESP_RX = D10 / GPIO18 (uart_rx_pin)

Underside pads 24–33 (MTDI, MTDO, EN, GND, MTMS, MTCK, BOOT, 3V3, BAT, GND) are not soldered and are omitted from this symbol. The PCB footprint marks them with no-via keep-outs.

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TX – OPEN-DRAIN LINE DRIVER

GPIO21 high turns Q2 on and pulls RED low (inverted UART TX). R1 holds Q2 off while the ESP32 is in reset or unpowered.

RX – INVERTING LEVEL SHIFTER

RED high turns Q1 on and pulls ESP_RX low (inverted UART RX). R5 is required: gdolib does not enable the internal pull-up.

GND INTERFACE – SECURITY+ 2.0

RED/CTRL

D1
CDSOD323-T15S

R4
100k

J1
2
1
RED
WHITE

250-1402

GND

GND

GND

J1 lands on the opener's wall-control terminals:
RED – Security+ 2.0 serial data (about 12.7 V idle)
WHITE – ground
D1 clamps transients on the line (15 V standoff).
R4 is a weak bias that defines the line when idle.
Obstruction status comes from the serial protocol;
the black obstruction wire is not used.

1. Firmware: ESPHome `seclplus_gdo` (`gdolib`) with `invert_uart = true`; `uart_tx_pin = GPIO21`, `uart_rx_pin = GPIO18`.
2. Both FET stages invert the signal; the firmware's UART inversion restores the logic sense.
3. Q1's gate sees the RED line directly: use a part rated ± 20 V Vgs (do not substitute A03400A, ± 12 V).
4. Ordering data: Manufacturer / MPN / Diq-Key PN symbol fields and `c6gdo-bom.csv`.

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